

Digital Twins in Industry 5.0: Challenges in Modeling and Communication

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Abstract—Digital Twins help to optimize the overall engineering process – from requirement setting through design and operations to maintenance and evolution. In order to handle the complexity they may bring throughout the lifecycle, it is worth modeling the behavior and interactions of physical systems. The same applies to digital twins, although modeling their interaction could be more challenging than handling the interaction of merely physical systems. Furthermore, the newly emerging industrial requirements on interoperability, reconfigurability, man-machine collaboration, and self-optimizing production necessitate the application of new networking paradigms. This paper provides an overview of the expectations and feasibility of modeling, communication, and networking challenges of digital twins for the upcoming revolution of Industry 5.0.

Index Terms—digital twins, system modeling, cloud robotics, private cellular networks, network slicing, industry 5.0

I. INTRODUCTION

The next wave of industrial (r)evolution is ahead of us. It is characterized by certain requirements and opportunities, including increased efficiency, improved human-machine cooperation, new communication paradigms, new models of information sharing and data representation, production value networks, and among others: Digital Twins (DTs). Although the idea of digital twins has been around for a while already, its implications on Cyber-Physical System of Systems have not been fully uncovered. Modeling for DTs is important both at design-time and run-time, although its various aspects – including high-fidelity, simulation, data representation, security, and interconnection – have a critical effect on industrial requirements, such as QoS, reliability, or connectivity.

The objective of this article is to present an overview of these requirements, challenges and opportunities from the perspective of Industry 5.0. Besides covering the issues around DTs, our paper provides an overview of the challenges of the next digital transformation, together with the networking-related requirements and opportunities, such as cloud capabilities, or slicing in private cellular networks.

In Section II of this paper, we discuss the modeling and interconnection aspects of DTs. Section III puts this in the perspective of the digital transformation of Industry 5.0. Section IV discusses the communication-related challenges, for which Section V presents certain key enablers in 5G and beyond cellular networks. Section VI concludes the paper with some lessons learned, and research gaps to cover.

II. MODELING AND INTERCONNECTION OF DIGITAL TWINS

A. What is a Digital Twin?

In general, DTs (Fig. 1) operate as a virtual representation of a real system that usually interacts with its environment. Many forms of a DT rely on domain and expert knowledge (background knowledge) and interact with third-party DTs, services, and data sources. It is impossible to give a single, ultimately clear definition of DTs due to the variety of their use cases, their place in complex systems, and the many technologies used to create them. In [1], the authors have evaluated definitions from papers, and they found that there are three key aspects of DTs that appear in many of the definitions. Namely, data-related, fidelity modeling-related, and model simulation-related aspects.

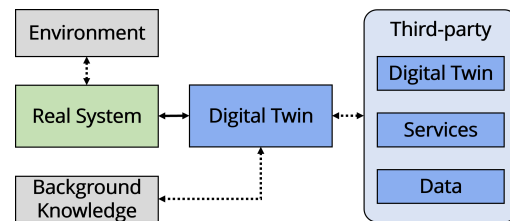


Fig. 1. Digital Twin and its environment

1) *Data aspect*: Data is one of the foundations of DTs. DTs can collect data from multiple sources, which can be quite far from the location of the DT. As data processing may even be required in real-time, not only advanced networking technologies (5G, 6G [2]) but also application-level communication solutions (e.g., DDS, MQTT, OPC UA) are required. Choosing the right data storage solution to support data processing is also necessary. Nowadays, there are plenty of solutions for managing and storing big data. However, in the case of DTs, special attention needs to be paid to the management of time-series data, and the semantic interpretation of data and the model [3, 4]. However, the findings in [3] show that knowledge and data-based solutions are much less used. The semantic management of data and models is supported by different knowledge-based methods (e.g., knowledge graphs[5]), or an ontology-based (e.g., [6]) data model. They can be built alongside DTs that support

both the validation of data integrity and efficient processing. Furthermore, a proper semantic representation helps in the interconnection of DTs.

2) *High-fidelity model aspect*: Just as in classical model-based system engineering (MBSE), in the case of DTs, models can and should be differentiated according to their goals. At a high abstraction level, a distinction can be made between behavioral (e.g., state machines) and structure-based (e.g., architecture) models. When designing DTs, we want to create a model that is as faithful as possible. However, this all-encompassing modeling is not possible for complex systems. One solution is to reduce the complexity of the model. In [7], the authors proposed a novel modeling method for complex DTs in multi-scale, multi-dimensional information and multi-scenarios. Building the right DT is also part of the fidelity aspect [8].

3) *Model simulation aspect*: The third analytical aspect of DTs is simulation [9]. The simulation of DTs can have several purposes. One is to predict the future behavior of the system based on real-time data from the original system. Furthermore, the simulation also provides the opportunity to perform the validation of actuation operations on the original system. During the simulation, the DT can check out several scenarios and select the best one based on optimization criteria [10, 11]. However, since we cannot create a completely faithful model of the system, there may be uncertainties [12, 13] in the simulation due to the abstraction.

B. Modeling of Digital Twins

In the modeling of DTs, two main modeling approaches can be distinguished [14]: (1) Behavior Modeling, which aims to describe the operation of systems, including the internal behavior of components and the interaction between them; (2) physical and geometric structural models, which can be used to model physical objects and their properties in the physical world. The emerging use of machine learning (ML) and artificial intelligence (AI) solutions [15] in the behavioral modeling of DTs may cause some concern [16]. For example, in the case of black-box models, the internal operation is not fully known; therefore, the results may not be sufficiently detailed and hard to be validated. Nowadays, the research on explainable artificial intelligence (xAI) can facilitate the integration of complex AI applications in critical environments [17].

The model lifecycle and management of the digital twin models [18] are very similar to the classic model lifecycle (MLC)[19]. However, besides the model construction, special attention has to be paid to (1) Assembly (e.g., in cyber-physical systems, modeling both the physical and the logical components); (2) Fusion (e.g., managing information from multiple sources [20]); (3) Verification (e.g., formal verification and validation in critical cases); (4) Modification (e.g., frequently changing environment); (5) Management (e.g., tracking changes).

C. Interconnection of Digital Twins

The Industrial Internet Consortium in [21] proposed integrational aspects that a DT architecture should support in order to achieve a fully integrated and interoperable DT. As asset management is a further challenge for DTs, in [22], the Industrial Internet Consortium (IIC) and Plattform Industrie 4.0 explored the connection of Asset Administration Shell (AAS) and DTs. The AAS approach is for managing and exchanging information about physical assets in the context of Industry 4.0. In addition to the typical integrational aspects, trust [23, 24] and security [25] also play an important role in a multi-party system. If the DT also uses data from external third-party data sources or integrates a service, the authenticity of the information must somehow be assured. Digital Twin Consortium proposed a novel dynamic trust management approach [23] between DT-based ecosystems. It is based on connection profiles, where parties can define their required and requested trust profiles for services. If the profiles match, they can use the services. From the security point of view, over the security measures of automated System of Systems[26], the EU also realized the emerging need for proper cyber security measures. This way, they presented a proposal for a regulation on cybersecurity requirements for products with digital elements called the Cyber Resilience Act [27].

III. FOSTERING THE NEW DIGITAL TRANSFORMATION

As Industry 4.0 emerged, it paved the way for the spreading of collaborative robots - created by many yet not recognized companies - not only in manufacturing plants but in research facilities also. Because of their accessibility, these robots got into the spotlight and thus appeared in more and more contexts by promoting new features and applications. Alongside physical robots, digital twin applications also appear on shopfloors to bring customers closer to manufacturing [28, 29].

Industrial users usually demand performance guarantees. It is no different from the relationship between networks and mission-critical industrial applications. These guarantees extend to system characteristics such as throughput, network latency or jitter [31, 32]. Because of the nature of 802.11 standards, QoS capabilities are limited thus industrial applications challenge WLANs. The vision of Industry 5.0 (I5.0) promotes new features which require versatile connectivity from cellular networks. I5.0 removes the physical limitations and builds virtual connectivity and capabilities that will challenge networking. By exploiting the potential of cellular networks, a whole new intelligent and agile realization of smart manufacturing becomes available. The key emphasis is on introducing the advancements of 5G networks with examples from real-world scenarios and examining what made cellular networks capable of being a pillar provider of wireless connectivity in industrial transformations. Fig. 2 presents the main characteristics of 5G verticals sectors.

IV. CHALLENGES OF DIGITAL TRANSFORMATION

Combining the advantages of collaborative robots and cellular networks, a hot topic emerges under the umbrella of smart

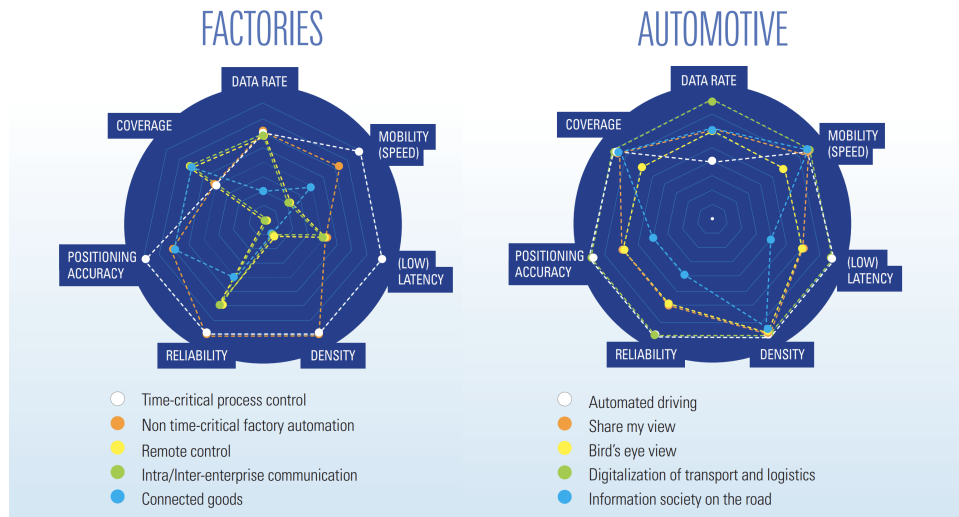


Fig. 2. Capabilities and requirements of verticals [30]

manufacturing, known as cloud robotics. The concept is to take computing-intensive tasks from the local device and place it on the network side. The main advantages and benefits of cloud robotics are the following:

- Shared and unified knowledge database
- Offloading computational-intensive tasks
- Lightweight devices
- Information sharing & synchronization among devices
- Machine to Machine (M2M) and Machine to Cloud (M2C) communication

With the advent of Industry 4.0, Digital Twins are also empowering industrial transformations. These virtual representations of physical assets and processes allow real-time monitoring, predictive maintenance, enhanced decision making and more. To address these enablers, networking background shall attempt - through coordination of the requests - to ensure the best possible matching of requirements [4].

1) *Quality of Service*: During the last decade, a lot of effort has been put into the improvement of QoS support in WLANs [33]. 802.11e standard - which is approved in 2005 - defines a set of QoS methods for wireless LAN applications by modifying the Medium Access Control (MAC) layer. Nowadays the most improved standard is the IEEE 802.11 ax, also known as WiFi 6 [34]. 802.11 standards have come a long way since the first wireless protocol is defined in 1997 yet still do not satisfy certain requirements to be accepted by industrial sectors. Through the more advanced QoS features in mobile networks, it shifts the focus from 802.11 standards to the 5th generation mobile communication network and its enabler features.

2) *Latency*: To properly use robotic devices, the network has to fulfill all the requirements of cloud robotics. When a robot relies almost fully on the cloud, losing connection would leave the device without control. The ISO Standard 10218-2 [35] is designed as a set of guidelines, inter alia,

communications and collaboration. The crucial parameters of known robots regarding connectivity are collected in Table I.

TABLE I
COMMUNICATION CAPABILITIES OF PREVALENT INDUSTRIAL ROBOTS

Device	UR3-e	MiR 100	KR C4	HEBI X-series
Control frequency	500 Hz	50 Hz	N/A	100 Hz-1 kHz
Ethernet/IP	X	X	X	X
MODBUS	X	TCP	TCP	-
PROFINET	X	-	X	-
Source	[36]	[37]	[38]	[39]

Important to note that the control frequency values in Table I are internal control loop frequencies. This does not necessarily mean that external commands from e.g., Ethernet has to be given at that rate. Most of these robots can handle external commands at much slower rates, however, when the whole control is meant to be placed in the cloud, parameters have to be obeyed. This could require even 1-millisecond latency that current 5G releases do not offer.

Besides control, DT tasks also demand low latency: to support predictive maintenance, synchronization, real-time data collection and processing are essential for digital twins. With the actual 5G release the latency is around 10 ms thus 100 Hz is sufficient for certain control loops. Later 3GPP releases are promising in reducing latency but it still requires time to offer this capability.

3) *Reliability*: Packet loss or data corruption during transmission is a crucial part of industrial communication. Although retransmitted data will not be an overhead for an average user, it can compromise complex processes in industrial systems [35]. For DTs, large amounts of data and real-time monitoring are required to provide predictive maintenance and prevent failures in advance. Therefore, fail-safe network reliability is often a crucial requirement.

4) *Connectivity*: As Table I shows, Ethernet is the most prevalent in the question of connectivity. MODBUS protocol - published by Schneider Electric [40] - is still being used by industrial partners, originally designed for PLC (Programmable Logic Controllers). It is a protocol ported to a multitude of physical buses with different implementations. The most common protocols are MODBUS RTU and MODBUS TCP supported by many partners ensuring interoperability.

Although MODBUS and Profinet are still present technologies [41], the spreading of Ethernet-based realizations is expected to take the leading role, in particular, with the promising advantages of mobile networks. The usage of standardized and more popular connection schemes encourages the spreading of smart solutions for industrial devices and their digital twin.

V. KEY ENABLERS OF CELLULAR NETWORKS

Industry 4.0 become the most important vertical of the 5th-generation cellular mobile network. Not only being accepted as an industrial wireless standard, but the enabling features made 5G the backbone of Industry 4.0 communication. The availability of 5G is started with the 3GPP Release 15 [42], initially emphasising ultra-reliable and low-latency communication (URLLC), Mobile Broadband (eMBB) and massive machine-type communication (MTC). By following the guidelines of ITU-R [43] the main characteristics of 5G such as wide-area coverage, deep indoor penetration, E2E latency up to 1 ms and reliability of 99,999% of packets, it becomes a proper solution to provide wireless connectivity for the transformation of manufacturing [44]. The state-of-the-art realizations related to the enablers are collected in Table II.

TABLE II

COLLECTION OF STATE-OF-THE-ART PUBLICATIONS IN RELEVANT TOPICS

Topic	Relevant findings
Cloud communication	[45, 46, 47]
Private cellular networks	[48, 49, 50]
Smart manufacturing	[51, 52, 53]
Industry 5.0 outlook	[54, 55, 56]

1) *Network Slicing*: This is one of the most awaited options of the new generation mobile network. A 5G slice supports different communication configurations by managing the separated control plane (CP) and user plane (UP). A slice contains 5G network functions and special radio access technology (RAT) settings. However, each slice can be configured with different parameters. One can consist of dedicated transport, radio and core resources including a dedicated UP function at the edge. Another network slice instance can share radio and transport resources between tenants, providing core network functions per tenant. These slices then flow as aggregated traffic to the core in the common infrastructure [57].

The advantage of network slicing is making mobile networks more agile, flexible, and versatile. Various sensors of robotic systems generate multiple data streams where each source deals with a different requirement for the network

like data rate, robustness or latency. With virtualization, slices can provide logical separation of different services and allow different groups to manage and implement changes. Enterprise customers can tailor network characteristics according to their use cases, and effectively trim unnecessary functions.

Challenges are also involved in network slicing. Without standalone 5G networks, the end-to-end virtualized network could not be reached. There is also scepticism regarding the security of the network slices [58]. Slice interdependence is also a relevant question due to finite resources, computing power or network functions. When a slice is overloaded, it can override one's ability to operate normal security protocols and leave it vulnerable.

2) *Private Cellular Networks*: The advantages of private mobile networks - also referred to as non-public networks - are reflected especially in smart manufacturing and its use cases. Companies in many industrial fields are deploying private cellular networks for their requests as an industry vertical [59]. Compared to other technologies like WiFi or the previous generation of cellular networks, besides the obvious improvements such as reduced latency, better performance and robustness, an important difference is that 5G networks require a higher initial capital investment than WiFi and other networks but in the long run, maintenance and management costs are significantly lower than any other technology.

3) *5G Cell Design*: According to current trends, mobile data consumption is exploding, caused by the spreading of smart devices and IoT. A less-mentioned fact is that more than 70% of this traffic takes place indoors such as in homes, offices, stations, malls and other public places [60].

TABLE III

PARAMETERS OF DIFFERENT 5G BASE STATIONS [61].

Cell type	Output power [W]	Cell radius [km]	Users	Location
Macrocell	10 <	8 - 30	2000 <	Outdoor
Microcell	1 - 10	0.2 - 2	100 - 2000	Indoor / Outdoor
Picocell	0.25 - 1	0.1 - 0.2	30 - 100	Indoor / Outdoor
Femtocell	0.001 - 0.25	0.01 - 0.1	1 - 30	Indoor

The increased use of radio frequency (RF) fields above 6 GHz - particularly for the 5G mobile network - has given rise to more and smaller radio cells. With its small physical size it can be mounted onto light poles rather than high buildings or towers. This is also necessary because higher frequency waves cannot cover as wide area as previous generation networks with lower frequency ranges, especially in warehouses full of obstructions. Therefore smaller cells are introduced which parameters are collected into Table III.

4) *Industry 5.0 Outlook*: Meanwhile Industry 4.0 aimed to make a step forward in production, connectivity and automation, Industry 5.0 is targeted at societal transitions by bringing people closer to manufacturing phases. One of the foreseeable hubs of the next industrial revolution is the wellbeing of workers which needs considerable investment. The vision beyond

productivity and efficiency is to reinforce the contribution of industry to society. The next industrial revolution places quality of life into the spotlight and requires a new, integrated approach to production, planning and consumption meanwhile managing and conserving natural resources.

VI. CONCLUSION

Although DTs are being used in more and more application domains, there are few solutions that take into account the recommendations related to the modelling aspects and interconnections. Therefore the ad-hoc solutions that are currently in use are not sustainable in the long term and are difficult to integrate into other systems and DT environments. This may also be due to the fact that beyond the recommendations, a standard to support mature integration cannot be discovered.

The two main challenges regarding DTs are (i) updating the status of the DT quickly based on the real-time status of the physical system, and (ii) integrating multiple DTs to construct a complex interconnected DT [8]. The former requires precise sensors and appropriately fast communication. The latter requires to deal with the challenges of data interoperability. Furthermore, the interconnection of DTs can lead to several challenges: (1) It opens up a larger cyber-attack surface, (2) trust between the parties, (3) integration and interpretation of third-party data.

Industrial revolutions are forming the way we think of warehouses, and now communication got an essential role since Industry 4.0. Devices are connected typically by wire, but wireless connection is a strongly spreading way of communication. With the application of private cellular networks in Industry 4.0 and further, Industry 5.0, a wide range of options become available for practical deployments.

Contrary to expectations, robots will not replace every industrial worker. However, dull, repetitive, and, more especially, dangerous tasks are steadily taken over by robots not to risk human life. The key aspect of the next industrial transformation is adapting a human-centric approach and improving the digital skills of workers to keep up. As cellular networks become the pillar of the industrial revolution, the forthcoming standards have to provide ways of advancement. However, the development of 5G and beyond mobile communication systems, and their related applications are still being formed and customized for various use cases based on the evolving research areas.

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